

Deployment Steps

1. Clone the Repository

```
ubuntu@ip-172-31-15-15:~$ git clone https://github.com/aditya8112soni-cr/eks-nginx-demo.git
Cloning into 'eks-nginx-demo'...
remote: Enumerating objects: 24, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (22/22), done.
remote: Total 24 (delta 6), reused 17 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (24/24), 11.40 KiB | 1.27 MiB/s, done.
Resolving deltas: 100% (6/6), done.
```

2. Navigate to the Terraform directory and initialise it

```
ubuntu@ip-172-31-15-156:~$ ls
eks-nginx-demo
ubuntu@ip-172-31-15-156:~$ cd eks-nginx-demo/
ubuntu@ip-172-31-15-156:~/eks-nginx-demo$ ls
README.md  kubernetes  terraform
ubuntu@ip-172-31-15-156:~/eks-nginx-demo$ cd terraform/
ubuntu@ip-172-31-15-156:~/eks-nginx-demo/terraform$ ls
eks.tf  main.tf  outputs.tf  provider.tf  variables.tf
ubuntu@ip-172-31-15-156:~/eks-nginx-demo/terraform$ terraform init
Initializing the backend...

Initializing modules...
Downloading registry.terraform.io/terraform-aws-modules/eks/aws 20.37.2 for eks...
- eks in .terraform/modules/eks
Downloading registry.terraform.io/terraform-aws-modules/vpc/aws 5.21.0 for vpc...
- vpc in .terraform/modules/vpc
eks_forget_profile in terraform/modules/eks/modules/forget_profile
```

3. After checking the plan , apply the terraform changes

```
# module.eks.module.eks_managed_node_group["default"].module.user_data.null_resource.validate_cluster_service_cidr will be created
+ resource "null_resource" "validate_cluster_service_cidr" {
+   id = (known after apply)
+ }

Plan: 57 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ cluster_certificate_authority_data = (sensitive value)
+ cluster_endpoint                 = (known after apply)
+ cluster_name                     = "nginx-demo-eks"
+ cluster_version                  = "1.36"
+ configure_kubectl                = "aws eks update-kubeconfig --region ap-south-1 --name nginx-demo-eks"
+ private_subnet_ids              = [
+   (known after apply),
+   (known after apply),
+ ]
+ public_subnet_ids               = [
+   (known after apply),
+   (known after apply),
+ ]
+ region                          = "ap-south-1"
+ vpc_id                          = (known after apply)
```

```
Outputs:

cluster_certificate_authority_data = <sensitive>
cluster_endpoint = "https://601c434aea1caa2efe739ca1a5cb6cf9.gr7.ap-south-1.eks.amazonaws.com"
cluster_name = "nginx-demo-eks"
cluster_version = "1.36"
configure_kubectl = "aws eks update-kubeconfig --region ap-south-1 --name nginx-demo-eks"
private_subnet_ids = [
  "subnet-00ca8e9e51c1ff5e9",
  "subnet-0a18eda4d347402eb",
]
public_subnet_ids = [
  "subnet-08a7bf4e22a16dedd",
  "subnet-02f91c034defcd73e",
]
region = "ap-south-1"
vpc_id = "vpc-0689239bbf67eee47"
```

3. Configure kubectl to connect to the cluster

```
Last login: Thu Sep 17 09:23:46 2026 from 223.181.40.92
[ubuntu@ip-172-31-15-156:~]$ aws eks update-kubeconfig --region ap-south-1 --name nginx-demo-eks
Updated context arn:aws:eks:ap-south-1:974669690983:cluster/nginx-demo-eks in /home/ubuntu/.kube/config
```

4. Navigate to the Kubernetes directory and verify nodes and create namespace , deploy nginx and create the service .

```
Updated context arn:aws:eks:ap-south-1:974669690983:cluster/nginx-demo-eks in /home/ubuntu/.kube/config
[ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
ip-10-0-1-4.ap-south-1.compute.internal Ready    <none>    3m23s   v1.36.3-eks-cb19647
ip-10-0-2-207.ap-south-1.compute.internal Ready    <none>    3m23s   v1.36.3-eks-cb19647
[ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl apply -f namespace.yaml
namespace/nginx-demo created
[ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
[ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl apply -f service.yaml
service/nginx-service created
[ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ ]
```

5. Once created validate them

```

service/nginx-service created
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl get deployment -n nginx-demo
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 2/2     2            2           35s
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl get pods -n nginx-demo
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-75d45597c7-ddb5h 1/1     Running   0          43s
nginx-deployment-75d45597c7-g6h9n 1/1     Running   0          43s
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl get pods -n nginx-demo -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES
nginx-deployment-75d45597c7-ddb5h 1/1     Running   0          53s   10.0.2.159   ip-10-0-2-207.ap-south-1.compute.internal <none>          <none>
nginx-deployment-75d45597c7-g6h9n 1/1     Running   0          53s   10.0.1.145   ip-10-0-1-4.ap-south-1.compute.internal <none>          <none>
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl get svc -n nginx-demo
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
nginx-service NodePort    172.20.155.197 <none>        80:30080/TCP 2m27s
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl get endpoints nginx-service -n nginx-demo
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME          ENDPOINTS          AGE
nginx-service 10.0.1.145:80,10.0.2.159:80 3m27s
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$

```

5. Allow port 8080 in the admin ec2 instance security group via my ip and port forward to 8080

i-0bcf8b421e9502a59 (eks-terraform-admin)

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security gr
-	sgr-0d572ede59768ce64	8080	TCP	223.181.40.92/32	launch-wiz:
-	sgr-0f29776dee383cffb	22	TCP	0.0.0.0/0	launch-wiz:

▼ Outbound rules

Filter rules

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```

/ # exit
ubuntu@ip-172-31-15-156:~/work/eks-nginx-demo/kubernetes$ kubectl port-forward \
-n nginx-demo \
svc/nginx-service \
8080:80 \
--address 0.0.0.0
Forwarding from 0.0.0.0:8080 -> 80
Handling connection for 8080
Handling connection for 8080
Handling connection for 8080
Handling connection for 8080

```

7. Check the NGINX UI via your ec2 public ip and port 8080

Click to go forward, hold to see history 13.204.79.45:8080

Gmail YouTube Maps Remote_Support

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](#).
Commercial support is available at [nginx.com](#).

Thank you for using nginx.

Relaunch to update

Why did we opt for public EC2 IP + port 8080, with kubectl port-forward to the Nginx Kubernetes Service instead of loadbalancer ??

Since my AWS account does not support load balancer creation at the moment . I have raised a support case with AWS and then have reverted as I dont have enough billing history , it can take 48-72 hours for this restriction to be uplifted .

The screenshot displays the AWS Support console interface. The browser address bar shows the URL: `support.console.aws.amazon.com/support/home#/case?displayId=178959348000788&language=en`. The page title is "Phone: AWS account cannot create Network Load Balancer".

The left sidebar contains the "Support Center" navigation menu with sections: "Your resources" (Support Dashboard, Support cases, Support Authorization), "AWS Support Plans" (Manage Support Plan, Business Support, Enterprise Support, Unified Operations), "AWS Support App in Slack" (Getting started), "Support Tools" (Compute Optimizer, AWS Health Dashboard, Trusted Advisor), and "AWS Knowledge resources" (Get started with AWS Support, Knowledge Center, AWS Documentation, rePost, Training and Certification).

The main content area is divided into sections: "Case details" (Basic information about this support case), "Support interactions" (Personalized recommendations for your issue), and "Correspondence".

The "Correspondence" section shows a conversation between AWS and Aditya. The AWS message, dated Sep 17, 2024, explains that the account lacks sufficient billing history for auto-approval of Elastic Load Balancers (ELB) and that a review may take 48-72 hours. The Aditya response, also dated Sep 17, 2024, states that he is deploying an Amazon EKS cluster in the ap-south-1 region and is unable to create a Network Load Balancer. He mentions that the Kubernetes service shows an "OperationNotPermitted" error and that he has tested creating a Network Load Balancer directly using the AWS CLI, which also resulted in the same error. He concludes by asking if there is an account-level restriction preventing the creation of Elastic Load Balancers and how it can be enabled.